

Martin Birkenbaum

SENIOR SOFTWARE ENGINEER

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Senior Software Engineer with 8+ years of experience designing, building, and deploying large-scale, distributed backend systems and high-throughput microservices. Expertise in cloud infrastructure, GitOps-driven automation, system design, and enhancing developer productivity through robust testing, advanced tooling, and clean code principles.

SKILLS

Languages:	Python, TypeScript, JavaScript, SQL
Technologies & Frameworks:	Docker, Kubernetes, Git, GitOps, PostgreSQL, pytest, Distributed Systems, Cloud Computing, System Design, Large-Scale Architecture, Performance Tuning, Debugging, CI/CD, Data Structures, Algorithms

WORK EXPERIENCE

Senior Software Engineer at **Initech Corporation** 2021-03-01 – 2026-07-01

Led architecture and development of high-throughput microservices and GitOps-driven infrastructure pipelines.

- Designed and implemented Python-based event-driven microservices, operating at massive scale to process 10M+ daily events within a distributed system architecture.
- Standardized highly reliable containerized deployments across large-scale infrastructure utilizing Docker, Kubernetes, and GitOps workflows, ensuring operational efficiency.
- Mentored 6 junior/mid-level engineers, fostering best practices for code quality, system health, and comprehensive testing standards.
- Developed custom internal developer toolchains and automation that significantly reduced CI pipeline execution times by 40%, enhancing engineering productivity and release velocity.

Software Engineer at **Hooli Inc.** 2018-06-01 – 2021-02-01

Backend engineer focused on building RESTful APIs, developer tooling, and automated deployment integrations.

- Developed robust REST APIs in Python/Django and TypeScript/Node.js, contributing system development code for critical backend services.
- Integrated third-party APIs and automated webhook processors, optimizing data flow and communication for external developer platforms.
- Implemented a comprehensive suite of unit and integration tests, raising test coverage from 60% to 92% and improving overall software test engineering practices and system reliability.

PROJECTS

JobGitOps (Python, Nix, GitHub Actions) 2026-06-01 – 2026-07-01

A GitOps-driven job application and tracking system utilizing GitHub Issues, Projects, and Actions.

- Developed python-jobsby scraper integration and two-pass LLM fit triage
- Designed reproducible local environments via Nix and devenv

EDUCATION

Bachelor of Science in Computer Science at **University of Washington**

2014-09-01 – 2018-06-01

GPA: 3.8